

Learning Activity Sheet for General Science

Quarter 2

Lesson

2

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Learning Activity Sheet for General Science Grade 11
Quarter 1: Lesson 2

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LEARNING ACTIVITY SHEET

Learning Area:	General Science	Quarter:	2
Lesson No.:	2	Date:	
Lesson Title/ Topic:	Application of Chemical Substances in Household and Personal Care Products		
Name:	Cherry V. Abrazado	Grade & Section:	11

I. Activity No. 1: Investigating the Cleaning Power of Household and Personal Care Products

II. Objective(s): At the end of the activity, you should be able

- Compare the effectiveness of different household and personal care cleaning agents.
 - Identify the active ingredients that influence stain removal.
 - Develop experimental and analytical skills through hands-on testing.
 - Test the pH level of products to understand their acidity or alkalinity.
- (Note: The learners may use litmus paper or any indicator for acidity and basicity.)

III. Materials Needed:

- Three types of cleaning agents (e.g., baking soda, bleach, detergent, shampoo, toothpaste)
- Water
- Cotton fabric or white tiles (as test surfaces)
- Cooking oil or a food-based stain (e.g., coffee, ink, grease)
- Small brushes or sponges
- Measuring cups
- pH strips for testing acidity or alkalinity

IV. Instructions: Analyze and solve each problem.

1. Prepare the Test Surfaces:
 - Apply equal amounts of oil or food stain to different fabric or tile surfaces.
2. Apply Cleaning Agents:
 - Add a small amount of each cleaning product to separate stained areas.
 - Ensure consistent application across all test samples.
3. Scrubbing & Rinsing:
 - Scrub each section using equal force and for the same duration.
 - Rinse the areas with water and observe results.
4. Compare Cleaning Effectiveness:
 - Examine which stain was removed most effectively.
 - Take notes on differences in cleaning power among products.
5. Test pH Levels:
 - Use pH strips to measure the acidity or alkalinity of each product.
 - Discuss how pH might impact cleaning effectiveness.

Cleaning Effectiveness and pH Testing Table

Product Name	Type of Stain Tested (Coffee/Ink/Grease)	Active Ingredients	Effectiveness (Strong/Moderate/Weak)	Observed Differences in Cleaning Power	pH Level (change of color, red-aci)
Baking soda					
Bleach					
Detergent					
Shampoo					
Toothpaste					

V. Synthesis:

- Which product removes the stain most effectively?

- Which cleaning product was the most effective? Which product showed the weakest cleaning power?

- Were there any noticeable differences in how quickly or efficiently the stains were removed?

- How does pH level of each product affect the cleaning effectiveness?

- Are there similarities in the ingredients of the most effective cleaning agents?
